

Hitec University Taxila, Department of Computer Engineering

Lab Manual for Mobile Application Development

Lab-4 Multiple Activity and Intents

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1 Introduction

An Intent is a simple message object that is used to communicate between android components such as activities, content providers, broadcast receivers and services. Intents are also used to transfer data between activities. Android application components can connect to other Android applications. This connection is based on a task description represented by an Intent object. Intents are asynchronous messages which allow application components to request functionality from other Android components. Intents allow you to interact with components from the same applications as well as with components contributed by other applications. For example, an activity can start an external activity for taking a picture.

2 Activity Time boxing

Table 1: Activity Time Boxing

Task#	Activity Name	Activity time	Total Time
1	Introduction	60mins	60min
2	Walk Through Task	60mins	60min
3	Evaluation Task	60min	60min

3 Objectives of the experiment

- To get basic understanding regarding Intents
- To know about the intent
- To be able to use Intents in your app
- To get to know about the benefits related to intents

4 Concept Map

An Intent is a simple message object that is used to communicate between android components such as activities, content providers, broadcast receivers and services. Intents are also used to transfer data between activities. Android application components can connect to other Android applications. This connection is based on a task description represented by an Intent object. Intents are asynchronous messages which allow application components to request functionality from other Android components. Intents allow you to interact with components from the same applications as well as with components contributed by other applications.

4.1 Intents

It specifies which application will satisfy the intent, by supplying either the target app's package name or a fully qualified component class name. You'll typically use an explicit intent to start a component in your own app, because you know the class name of the activity or service you want to start. For example, you might start a new activity within your app in response to a user action or start a service to download a file in the background.

4.1.1 Syntax:

```
Intent I = new Intent(getApplicationContext(),NextActivity.class);  
I.putExtra("value1" , "This value for Next Activity");  
I.putExtra("value2" , "This value for Next Activity");
```

5 Walk through Task

5.1.1 Creating New Activity

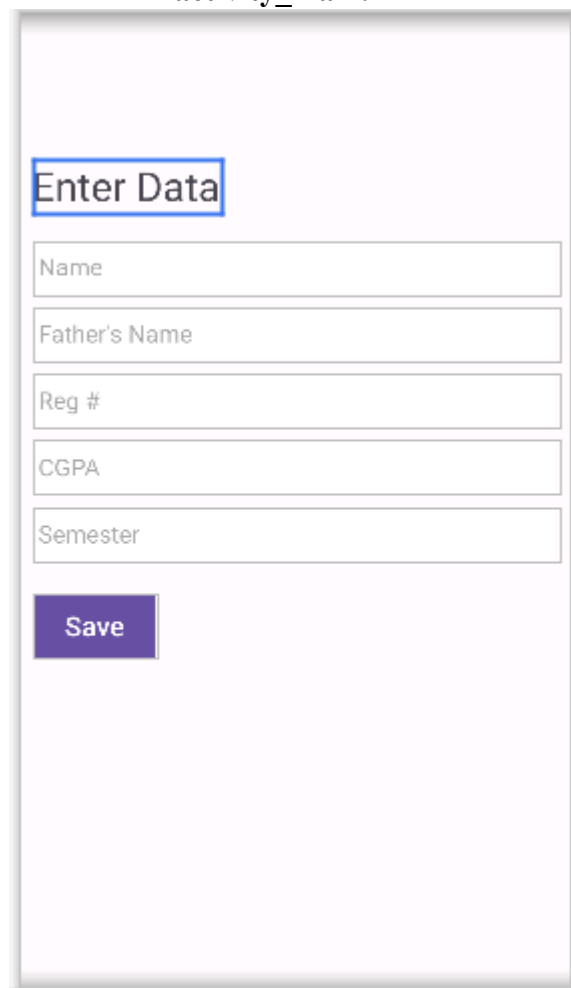
Step 1: Create Two Activities:

Create a new project just like you have created in your previous labs and create one more empty activity.

Step 2: Design XMLs of both Activities:

Designs of both activities are given below, just use the given xml code which is given in this step.

activity_main.xml



The screenshot displays a mobile application interface for data entry. At the top, the text "Enter Data" is enclosed in a blue rectangular box. Below this, there are five vertically stacked text input fields with the following labels: "Name", "Father's Name", "Reg #", "CGPA", and "Semester". At the bottom of the form, there is a purple button with the text "Save" in white.

Use the below code to design XMLs of both activates. Main Activity XML:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="10dp">

    <TextView
        android:id="@+id/tv_Home"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Enter Data"
        android:textSize="30sp"
        android:layout_marginTop="100dp"
        android:layout_marginBottom="20dp"
    />
    <EditText
        android:id="@+id/et_Name"
        android:layout_width="match_parent"
        android:layout_height="40dp"
        android:hint=" Name"
        android:background="#fff"
        android:lines="1"
    />
    <EditText
        android:id="@+id/et_FatherName"
        android:layout_width="match_parent"
        android:layout_height="40dp"
        android:background="#fff"
        android:layout_marginTop="10dp"
        android:hint=" Father's Name"
        android:lines="1"
    />
    <EditText
        android:id="@+id/et_Reg"
        android:layout_width="match_parent"
        android:layout_height="40dp"
        android:background="#fff"
        android:layout_marginTop="10dp"
        android:hint=" Reg #"
        android:lines="1"
    />
    <EditText
        android:id="@+id/et_CGPA"
        android:layout_width="match_parent"
        android:layout_height="40dp"
        android:background="#fff"
        android:layout_marginTop="10dp"
        android:hint=" CGPA"
        android:lines="1"
    />
    <EditText
        android:id="@+id/et_Semester"
        android:layout_width="match_parent"
        android:layout_height="40dp"

```

```

        android:background="#fff"
        android:layout_marginTop="10dp"
        android:hint=" Semester"
        android:lines="1"
    />
    <Button
        android:id="@+id/btn_Save"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="25dp"
        android:background="#fff"
        android:text="Save"
        android:textSize="20sp"
    />
</LinearLayout>

```

activity_next.xml

```

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="10dp"

```

```

        android:orientation="vertical">

        <TextView
            android:id="@+id/textoutput"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="50dp"
            android:text="Output Screen"
            android:textSize="30sp"
            android:textColor="#000"/>
        <TextView
            android:id="@+id/textviewName"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:textSize="20sp"
            android:text="Name"
            android:layout_marginTop="20dp"
            />
        <TextView
            android:id="@+id/textviewFatherName"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Father Name"
            android:textSize="20sp"
            android:layout_marginTop="20dp"
            />
        <TextView
            android:id="@+id/textviewReg"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Registration #"
            android:textSize="20sp"
            android:layout_marginTop="20dp"
            />
        <TextView
            android:id="@+id/textviewCGPA"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:textSize="20sp"
            android:text="CGPA"
            android:layout_marginTop="20dp"
            />
        <TextView
            android:id="@+id/textviewSemester"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Semester"
            android:textSize="20sp"
            android:layout_marginTop="20dp" />
    </LinearLayout>

```

Step 3: Initialize all the UI components:

The very first step of java code is to create objects of all the UI controls used in XML of both activities.

```

EditText etName, etFatherName, etReg, etCGPA, etSemester;
Button btnSave;

etName = (EditText) findViewById(R.id.et_Name);
etFatherName = (EditText) findViewById(R.id.et_FatherName) ;
etReg = (EditText) findViewById(R.id.et_Reg);
etCGPA = (EditText) findViewById(R.id.et_CGPA);
etSemester = (EditText) findViewById(R.id.et_Semester);
btnSave = (Button) findViewById(R.id.btn_Save);

```

Step 4: Create OnClickListener:

Create an OnClickListener for the Button object and implement the following code in it.

```

btnSave.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        String name, fname, reg, cgpa, semester;
        name = etName.getText().toString();
        fname = etFatherName.getText().toString();
        reg= etReg.getText().toString();
        cgpa = etCGPA.getText().toString();
        semester= etSemester.getText().toString();

        Intent intObj = new Intent( MainActivity.this,NextActivity.class);
        intObj.putExtra("USERNAME", name);
        intObj.putExtra("FATHERNAME", fname);
        intObj.putExtra("REG", reg);
        intObj.putExtra( "CGPA", cgpa);
        intObj.putExtra( "SEMESTER", semester);
        startActivity(intObj);
    }
});

```

Step 5: Get sent data from Main Activity using Intent:

To get the sent data in the next activity use Intent object and call getIntent() function. Next you have to save this data in String objects and then display them on TextViews using SetText() function.

```

public class NextActivity extends AppCompatActivity {
    TextView txtName, txtFatherName, txtReg, txtCGPA, txtSemester;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_next);

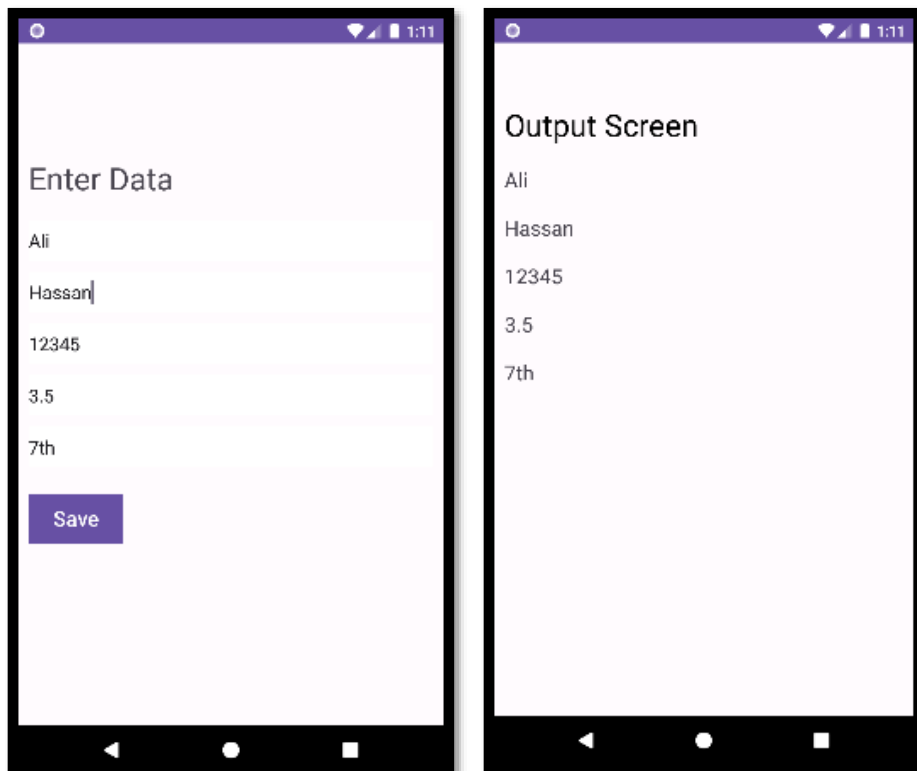
        txtName = (TextView) findViewById(R.id. textviewName) ;
        txtFatherName = (TextView) findViewById(R.id. textviewFatherName) ;
        txtReg = (TextView) findViewById(R.id. textviewReg) ;
        txtCGPA = (TextView) findViewById(R.id. textviewCGPA);
        txtSemester = (TextView) findViewById(R.id. textviewSemester) ;

        Intent getIntentObj = getIntent();
        String uname = (String) getIntentObj.getStringExtra("USERNAME");
        String fname = (String) getIntentObj.getStringExtra( "FATHERNAME");
        String reg = (String) getIntentObj.getStringExtra( "REG");
        String cgpa = (String) getIntentObj.getStringExtra( "CGPA");
        String semster = (String) getIntentObj.getStringExtra( "SEMESTER");
        txtName.setText (uname) ;
        txtFatherName. setText (fname) ;
        txtReg.setText (reg);
        txtCGPA. setText (cgpa);
        txtSemester.setText (semster) ;
    }
}

```

5.1.2 Step 6: Now Run Project:

Click “Run” or press “Shift + F10”.



6 Home Task

Design a simple login app to check entered username and password, if user entered wrong user name or password then show Toast message “You have entered wrong user name or password”, on successful entering the user name and password, bring to home page.

